Network flow models

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Learning objectives (1/2)

At the end of this chapter you will be able to:

- describe a network with balances, lower bounds, capacities and costs, and write the minimum cost flow model, also in matrix form with the incidence matrix;
- check the necessary conditions for the existence of a feasible flow, and build counterexamples;
- remove the lower bounds by a substitution, and balance an instance with a dummy node;

Learning objectives (2/2)

At the end of this chapter you will be able to:

- prove that the incidence matrix is totally unimodular, and deduce that the model has integer optimal solutions when the data are integer;
- write the dual with the potentials π_i, compute the reduced costs and certify the optimality of a flow;
- recognize the shortest path, the maximum flow, the transportation and the assignment problems as minimum cost flow models.

Prerequisites

- From this module, the chapter «Graphs»: directed graphs, the node–arc incidence matrix, spanning trees and the rank of the incidence matrix
- From the module *Linear Programming*, the chapters «Duality theory» and «The geometry of Linear Programming»: the dual of an LP, weak and strong duality, vertices of a polyhedron
- Determinants: the Laplace expansion along a column and Cramer's rule

Networks and flows

Introduction: Networks and flows

- We add data to a directed graph: a **balance** on every node, and a lower bound, a capacity and a cost on every arc
- A **flow** sends units along the arcs, so that every node sends out exactly its balance
- These are the objects of the whole part of the course on flows: we fix them on the **reference network**, which we use in every example of the chapter

Definition 1: network

*A **network** is a directed graph $G = (N, A)$, with $n = |N|$ nodes and $m = |A|$ arcs, together with the following data.*

- *Every node $i \in N$ has a **balance** $b_i \in \mathbb{R}$. The node is a **supply node** if $b_i > 0$, a **demand node** if $b_i < 0$, and a **transshipment node** if $b_i = 0$.*
- *Every arc $(i, j) \in A$ has a **lower bound** l_{ij} and a **capacity** u_{ij}, with $0 \leq l_{ij} \leq u_{ij} \leq +\infty$, and a **cost** $c_{ij} \in \mathbb{R}$ per unit of flow.*

When the lower bounds are not given, $l_{ij} = 0$ for every arc.

Definition 2: flow

*Let $G = (N, A)$ be a network. A **flow** is a vector $\boldsymbol{x} \in \mathbb{R}^m$ with one entry x_{ij}, the amount sent along the arc, for every $(i, j) \in A$.*

- *The flow satisfies the **flow conservation** constraint of node i if what leaves i minus what enters i equals the balance:*

$$\sum_{(i,j)\in\delta^+(i)} x_{ij} - \sum_{(j,i)\in\delta^-(i)} x_{ji} = b_i.$$

- *The flow is **feasible** if it satisfies flow conservation at every node and $l_{ij} \leq x_{ij} \leq u_{ij}$ for every arc. Its **cost** is $\sum_{(i,j)\in A} c_{ij}\, x_{ij}$.*

Reading the conservation constraints

- A supply node sends out b_i units more than it receives
- A demand node receives $-b_i$ units more than it sends
- A transshipment node passes on everything it receives

- With the node–arc incidence matrix $\boldsymbol{B}$ of the chapter «Graphs», flow conservation at all the nodes reads

$$\boldsymbol{B}\boldsymbol{x} = \boldsymbol{b}$$

- Row i of $\boldsymbol{B}\boldsymbol{x}$ is the outflow minus the inflow of i

Example: the reference network (1/3)

- We add data to the reference graph of the chapter «Graphs», with the arcs numbered $a_1, a_2, \ldots, a_9$ in lexicographic order
- The balances are

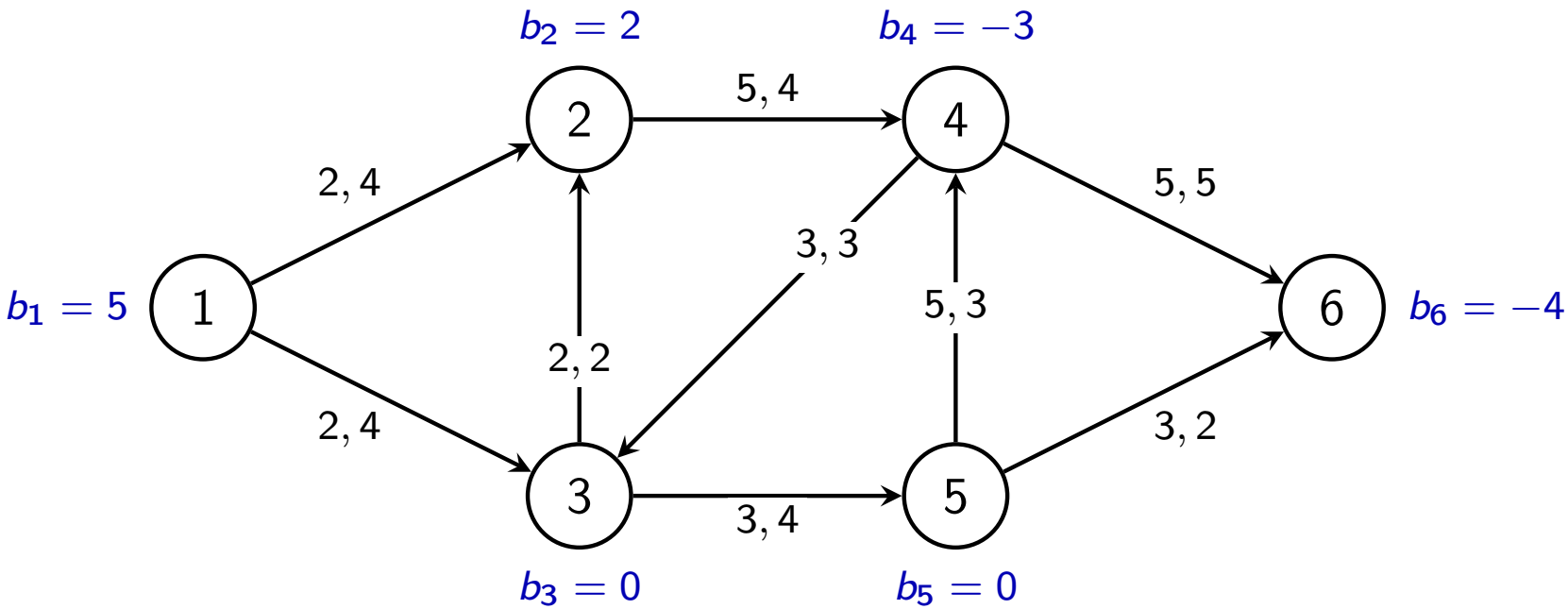
$$b_1 = 5, \qquad b_2 = 2, \qquad b_3 = 0,$$
$$b_4 = -3, \qquad b_5 = 0, \qquad b_6 = -4,$$

  so nodes 1 and 2 supply 7 units, nodes 4 and 6 demand them, and nodes 3 and 5 are transshipment nodes
- The lower bounds are zero, and the costs and capacities are:

arc	$(1,2)$	$(1,3)$	$(2,4)$	$(3,2)$	$(3,5)$	$(4,3)$	$(4,6)$	$(5,4)$	$(5,6)$
c_{ij}	2	2	5	2	3	3	5	5	3
u_{ij}	4	4	4	2	4	3	5	3	2

Example: the reference network (2/3)

In the figure each arc carries the pair c_{ij}, u_{ij}, and each node its balance b_i (in blue).



Example: the reference network (3/3)

- The flow $\bar{\boldsymbol{x}}$ with

$$\bar{x}_{12} = 1,\ \bar{x}_{13} = 4,\ \bar{x}_{24} = 3,\ \bar{x}_{32} = 0,\ \bar{x}_{35} = 4,$$
$$\bar{x}_{43} = 0,\ \bar{x}_{46} = 2,\ \bar{x}_{54} = 2,\ \bar{x}_{56} = 2$$

  is feasible

- For example, at node 4 the outflow is $\bar{x}_{43} + \bar{x}_{46} = 2$, the inflow is $\bar{x}_{24} + \bar{x}_{54} = 5$, and

$$2 - 5 = -3 = b_4$$

- The arcs $(1, 3)$, $(3, 5)$ and $(5, 6)$ are at capacity

- The cost of $\bar{\boldsymbol{x}}$ is 63

From nodes to sets of nodes

- Conservation holds node by node: we now look at what happens to a whole **set of nodes**
- For a set of nodes $S \subseteq N$, we denote by
  - $\delta^+(S)$ the arcs with the tail in S and the head outside S,
  - $\delta^-(S)$ the arcs with the head in S and the tail outside S,
- For a set of arcs F, we write

$$x(F) = \sum_{(i,j) \in F} x_{ij}$$

Lemma 1: flow across a cut

Let $\boldsymbol{x}$ *satisfy flow conservation at every node of a network* $G = (N, A)$*. Then, for every* $S \subseteq N$*,*

$$x(\delta^+(S)) - x(\delta^-(S)) = \sum_{i \in S} b_i.$$

- Idea of the proof: add up the conservation constraints of the nodes of S, and look at which arcs survive

Proof

- Add up the conservation constraints of the nodes of S; the right-hand side is $\sum_{i \in S} b_i$
- An arc with both endpoints in S appears twice, with $+x_{ij}$ in the row of its tail and $-x_{ij}$ in the row of its head: it cancels
- An arc of $\delta^+(S)$ appears only in the row of its tail, with $+x_{ij}$
- An arc of $\delta^-(S)$ appears only in the row of its head, with $-x_{ij}$
- The other arcs do not appear



Flow across a cut on the reference network

- In the reference network, the set $S = \{1, 2, 3\}$ has

$$\delta^+(S) = \{(2, 4), (3, 5)\}, \qquad \delta^-(S) = \{(4, 3)\}$$

- The flow $\bar{\boldsymbol{x}}$ gives

$$\bar{x}_{24} + \bar{x}_{35} - \bar{x}_{43} = 3 + 4 - 0 = 7 = b_1 + b_2 + b_3$$

Summary

Networks:

- balances on the nodes; lower bounds, capacities and costs on the arcs
- supply nodes ($b_i > 0$), demand nodes ($b_i < 0$), transshipment nodes ($b_i = 0$)

Flows:

- a feasible flow satisfies $\boldsymbol{Bx} = \boldsymbol{b}$ and the bounds
- across every cut, the net outflow equals the total balance of the nodes inside

The minimum cost flow model

Introduction: The minimum cost flow model

- Among all the feasible flows we look for the **cheapest one**
- The problem is a Linear Programming model with one variable per arc and one equation per node
- The special structure of its matrix, the incidence matrix, gives all the properties of this chapter

Definition 3: minimum cost flow problem

*Given a network $G = (N, A)$ with balances b_i, lower bounds l_{ij}, capacities u_{ij} and costs c_{ij}, the **minimum cost flow problem** asks for a feasible flow of minimum cost.*

- We use one continuous variable x_{ij} for every arc $(i, j) \in A$, the flow on the arc

The MCF model

Using these variables, a Linear Programming (LP) model for the minimum cost flow problem reads as follows:

$$\min \sum_{(i,j)\in A} c_{ij}\, x_{ij}$$

$$\text{subject to} \sum_{(i,j)\in\delta^+(i)} x_{ij} - \sum_{(j,i)\in\delta^-(i)} x_{ji} = b_i, \qquad \forall i \in N,$$

$$l_{ij} \le x_{ij} \le u_{ij}, \quad \forall (i,j) \in A.$$

This LP model is referred to as the MCF model.

Objective function and constraints

Description of the objective function and constraints of the MCF model:

- the linear objective function minimizes the total cost of the flow;
- the first constraints are the **flow conservation** constraints, one for every node;
- finally, the last constraints impose the lower bound and the capacity of every arc.

The MCF model in matrix form

- With the node–arc incidence matrix $\boldsymbol{B}$, the vector $\boldsymbol{c}$ of the costs and the vectors $\boldsymbol{l}$, $\boldsymbol{u}$ of the bounds, the MCF model is

$$\min \left\{ \boldsymbol{c}'\boldsymbol{x} \ : \ \boldsymbol{B}\boldsymbol{x} = \boldsymbol{b}, \ \boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u} \right\}$$

- The model has m variables and n equations
- The column of x_{ij} in $\boldsymbol{B}$ has only two nonzero entries, $+1$ in row i and -1 in row j
- This structure gives **all the properties** of this chapter

Example: the MCF model of the reference network (1/2)

For the reference network the MCF model has 9 variables and 6 equations:

$$
\begin{array}{lrrrrrrrrrrr}
\min & 2x_{12} & +2x_{13} & +5x_{24} & +2x_{32} & +3x_{35} & +3x_{43} & +5x_{46} & +5x_{54} & +3x_{56} & & \\
\text{subject to} & x_{12} & +x_{13} & & & & & & & & = & 5 \\
& -x_{12} & & +x_{24} & -x_{32} & & & & & & = & 2 \\
& & -x_{13} & & +x_{32} & +x_{35} & -x_{43} & & & & = & 0 \\
& & & -x_{24} & & & +x_{43} & +x_{46} & -x_{54} & & = & -3 \\
& & & & & -x_{35} & & & +x_{54} & +x_{56} & = & 0 \\
& & & & & & & -x_{46} & & -x_{56} & = & -4
\end{array}
$$

Example: the MCF model of the reference network (2/2)

Together with the bounds:

$$
\begin{aligned}
0 \leq x_{12} \leq 4,\\
0 \leq x_{13} \leq 4,\\
0 \leq x_{24} \leq 4,\\
0 \leq x_{32} \leq 2,\\
0 \leq x_{35} \leq 4,\\
0 \leq x_{43} \leq 3,\\
0 \leq x_{46} \leq 5,\\
0 \leq x_{54} \leq 3,\\
0 \leq x_{56} \leq 2.
\end{aligned}
$$

The coefficient matrix of the equations is the incidence matrix $\boldsymbol{B}$ of the reference graph.

Summary

- The minimum cost flow problem is the LP

$$\min\{\boldsymbol{c}'\boldsymbol{x} : \boldsymbol{B}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u}\}$$

- One variable per arc, one conservation equation per node
- Every column of $\boldsymbol{B}$ has exactly one $+1$ (the tail) and one -1 (the head)

Feasibility conditions

Introduction: Feasibility conditions

- Before looking for the cheapest flow, we ask whether a feasible flow **exists** at all
- The lemma on the flow across a cut gives two necessary conditions: the balance of the network and the cut condition
- We show with counterexamples that the balance alone is not enough

Definition 4: balanced network

A network is ***balanced*** *if* $\sum_{i \in N} b_i = 0$*, that is, if the total supply equals the total demand.*

Proposition 1: a feasible network is balanced

If a network $G = (N, A)$ *has a feasible flow, then* $\sum_{i \in N} b_i = 0$*.*

Proof

- Apply the lemma on the flow across a cut with $S = N$
- No arc leaves or enters N, so

$$\delta^+(N) = \delta^-(N) = \emptyset \qquad \text{and} \qquad \sum_{i \in N} b_i = 0$$

- In matrix form: the rows of $\boldsymbol{B}$ add up to the zero vector, hence so do the entries of $\boldsymbol{b} = \boldsymbol{B}\boldsymbol{x}$



Proposition 2: cut condition

If a network $G = (N, A)$ has a feasible flow, then for every $S \subseteq N$

$$\sum_{i \in S} b_i \leq u(\delta^+(S)) - l(\delta^-(S)),$$

where $u(F)$ and $l(F)$ are the sums of the capacities and of the lower bounds of the arcs of F.

Proof

- Let $\boldsymbol{x}$ be a feasible flow
- We use the lemma on the flow across a cut, the bounds $x_{ij} \leq u_{ij}$ on $\delta^+(S)$ and the bounds $x_{ij} \geq l_{ij}$ on $\delta^-(S)$:

$$\sum_{i \in S} b_i = x(\delta^+(S)) - x(\delta^-(S)) \leq u(\delta^+(S)) - l(\delta^-(S)).$$

□

Reading the cut condition

- The supply of the nodes of S, net of their demand, must be able to leave S through the arcs of $\delta^+(S)$
- The arcs of $\delta^-(S)$ bring at least their lower bounds into S
- A balanced network can still have **no feasible flow**: the balance is only a necessary condition

Example: balanced networks without a feasible flow (1/2)

- We keep the arcs, costs and capacities of the reference network and change the balances
- **First case.** With $b_5 = -1$, $b_6 = 1$ and $b_i = 0$ for the other nodes, the network is balanced
- But node 6 has no leaving arc: for $S = \{6\}$ we get $\delta^+(S) = \emptyset$
- The cut condition fails, since $b_6 = 1 > 0$

Example: balanced networks without a feasible flow (2/2)

- **Second case.** With $b_1 = 9$, $b_6 = -8$ and the other balances as in the reference network, the network is balanced, since

$$9 + 2 - 3 - 8 = 0$$

- For $S = \{1\}$ we get

$$u(\delta^+(S)) = u_{12} + u_{13} = 8 < 9 = b_1 :$$

node 1 cannot send out its supply

- For the reference network itself, all the 62 sets S with $\emptyset \neq S \neq N$ satisfy the cut condition, and the flow $\bar{\boldsymbol{x}}$ is feasible

Summary

- A network with a feasible flow is **balanced**
- Every cut can carry the net supply of the nodes inside it: the **cut condition**
- These conditions are **necessary**: a balanced network may have no feasible flow

Removing the lower bounds

Introduction: Removing the lower bounds

- Lower bounds force some flow on some arcs
- A change of variables moves them into the balances and the capacities, and leaves a network with **zero lower bounds**
- From then on, we can always assume $\boldsymbol{l} = 0$, as we do in the section on the dual model

Definition 5: substitution of the lower bounds

Given a network with lower bounds $\boldsymbol{l}$*, the* ***substituted network*** *has the same nodes, arcs and costs, zero lower bounds and*

$$b'_i = b_i - \sum_{(i,j)\in\delta^+(i)} l_{ij} + \sum_{(j,i)\in\delta^-(i)} l_{ji}, \qquad \forall i \in N,$$

$$u'_{ij} = u_{ij} - l_{ij}, \qquad \forall (i,j) \in A,$$

that is, $\boldsymbol{b}' = \boldsymbol{b} - \boldsymbol{B}\boldsymbol{l}$ *and* $\boldsymbol{u}' = \boldsymbol{u} - \boldsymbol{l}$*.*

Sending the lower bounds in advance

The l_{ij} units that must travel on (i, j) are sent in advance.

- The tail i has l_{ij} units less to send
- The head j has l_{ij} units less to receive
- The arc has l_{ij} units less of free capacity

Proposition 3: removing the lower bounds

The map $\boldsymbol{x}' = \boldsymbol{x} - \boldsymbol{l}$ *is a one-to-one correspondence between the feasible flows of a network and those of its substituted network, and*

$$\boldsymbol{c}'\boldsymbol{x} = \boldsymbol{c}'\boldsymbol{x}' + \boldsymbol{c}'\boldsymbol{l}.$$

Hence the two MCF *models have the same optimal solutions, up to the shift by* $\boldsymbol{l}$*, and their optimal values differ by the constant* $\boldsymbol{c}'\boldsymbol{l}$*.*

Proof

The map is linear and invertible, with inverse $\boldsymbol{x} = \boldsymbol{x}' + \boldsymbol{l}$. For every $\boldsymbol{x} \in \mathbb{R}^m$:

1. $\boldsymbol{B}\boldsymbol{x} = \boldsymbol{b}$ if and only if

$$\boldsymbol{B}\boldsymbol{x}' = \boldsymbol{B}\boldsymbol{x} - \boldsymbol{B}\boldsymbol{l} = \boldsymbol{b} - \boldsymbol{B}\boldsymbol{l} = \boldsymbol{b}';$$

2. $\boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u}$ if and only if $0 \leq \boldsymbol{x}' \leq \boldsymbol{u} - \boldsymbol{l} = \boldsymbol{u}'$;

3. $\boldsymbol{c}'\boldsymbol{x} = \boldsymbol{c}'(\boldsymbol{x}' + \boldsymbol{l}) = \boldsymbol{c}'\boldsymbol{x}' + \boldsymbol{c}'\boldsymbol{l}$.

The cost of every feasible flow changes by the same constant, so the optimal solutions correspond. □

Example: a lower bound on the reference network (1/2)

- In the reference network we impose $l_{32} = 1$: at least one unit must travel from 3 to 2
- The column of $(3, 2)$ in $\boldsymbol{B}$ has $+1$ in row 3 and -1 in row 2, so

$$\boldsymbol{B}\boldsymbol{l} = (0, -1, 1, 0, 0, 0)'$$

- Hence

$$\boldsymbol{b}' = \boldsymbol{b} - \boldsymbol{B}\boldsymbol{l} = (5,\ 3,\ -1,\ -3,\ 0,\ -4)', \qquad u'_{32} = 2 - 1 = 1$$

Example: a lower bound on the reference network (2/2)

- The constant in the objective is

$$\boldsymbol{c}'\boldsymbol{l} = c_{32}\, l_{32} = 2$$

- Node 3 becomes a demand node, because it has already sent one unit, and node 2 has one more unit to send
- The substituted network has optimal value 60
- Adding the constant 2, the optimal value with $l_{32} = 1$ is 62

Summary

- The substitution $\boldsymbol{x} = \boldsymbol{l} + \boldsymbol{x}'$ removes the lower bounds
- The new data are

$$\boldsymbol{b}' = \boldsymbol{b} - \boldsymbol{B}\boldsymbol{l}, \qquad \boldsymbol{u}' = \boldsymbol{u} - \boldsymbol{l}$$

- The objective changes by the constant $\boldsymbol{c}'\boldsymbol{l}$

Unbalanced instances

Introduction: Unbalanced instances

- In practice the total supply often **exceeds** the total demand: not all the supply is used
- Such an instance has no feasible flow with equalities, since it is not balanced
- A **dummy node** absorbs the excess and turns it into a balanced network

Definition 6: unbalanced instance

An instance with $\sum_{i \in N} b_i > 0$ *is* ***unbalanced****: the total supply exceeds the total demand. Every demand must be met, and every supply node* i *sends out at most* b_i *units:*

$$\sum_{(i,j) \in \delta^+(i)} x_{ij} - \sum_{(j,i) \in \delta^-(i)} x_{ji} \leq b_i, \quad \forall i \in N : \; b_i > 0,$$

$$\sum_{(i,j) \in \delta^+(i)} x_{ij} - \sum_{(j,i) \in \delta^-(i)} x_{ji} = b_i, \quad \forall i \in N : \; b_i \leq 0,$$

with the bounds $l_{ij} \leq x_{ij} \leq u_{ij}$ *and the objective* $\min \sum_{(i,j) \in A} c_{ij} \, x_{ij}$.

Definition 7: dummy node

Let $P = \{i \in N : b_i > 0\}$ *be the supply nodes and* $e = \sum_{i \in N} b_i > 0$ *the excess. The* ***balanced network*** *adds a* ***dummy node*** d *with balance* $b_d = -e$*, and an arc* (i, d) *for every* $i \in P$ *with* $l_{id} = 0$*,* $u_{id} = +\infty$ *and* $c_{id} = 0$*.*

Proposition 4: the dummy node

The feasible solutions of the unbalanced instance and the feasible flows of the balanced network correspond, with the same cost: a flow of the balanced network restricted to A is feasible for the unbalanced instance, and a solution of the unbalanced instance becomes a feasible flow with $x_{id} = b_i - (\text{outflow} - \text{inflow of } i)$.

Proof (1/2)

- For a vector $\boldsymbol{x}$ on A, let $s_i(\boldsymbol{x})$ be the outflow minus the inflow of node i along the arcs of A
- **From the balanced network.** Let $(\boldsymbol{x}, (x_{id})_{i \in P})$ be feasible for the balanced network
- At $i \in P$ conservation reads

$$s_i(\boldsymbol{x}) + x_{id} = b_i \quad \text{with } x_{id} \geq 0, \qquad \text{so } s_i(\boldsymbol{x}) \leq b_i$$

- At the other nodes of N it reads $s_i(\boldsymbol{x}) = b_i$

Proof (2/2)

- **To the balanced network.** Let $\boldsymbol{x}$ be feasible for the unbalanced instance, and set $x_{id} = b_i - s_i(\boldsymbol{x}) \geq 0$ for $i \in P$
- Conservation holds at every node of N, and at d the inflow is

$$\sum_{i \in P} x_{id} = \sum_{i \in P} b_i - \sum_{i \in N} s_i(\boldsymbol{x}) + \sum_{i \notin P} s_i(\boldsymbol{x})$$

$$= \sum_{i \in P} b_i - 0 + \sum_{i \notin P} b_i = e,$$

  since the values $s_i(\boldsymbol{x})$ add up to zero: every arc of A contributes $+x_{ij}$ to its tail and $-x_{ij}$ to its head

- Hence d receives exactly $-b_d = e$ units

The dummy arcs cost zero, so the two costs coincide. □

The dummy arcs

- The flow on (i, d) is the part of the supply of i that is **not used**
- If the total demand exceeds the total supply and the demands are upper limits, a dummy node d with balance

$$b_d = -\sum_{i \in N} b_i$$

and arcs (d, j) to the demand nodes balances the network in the same way

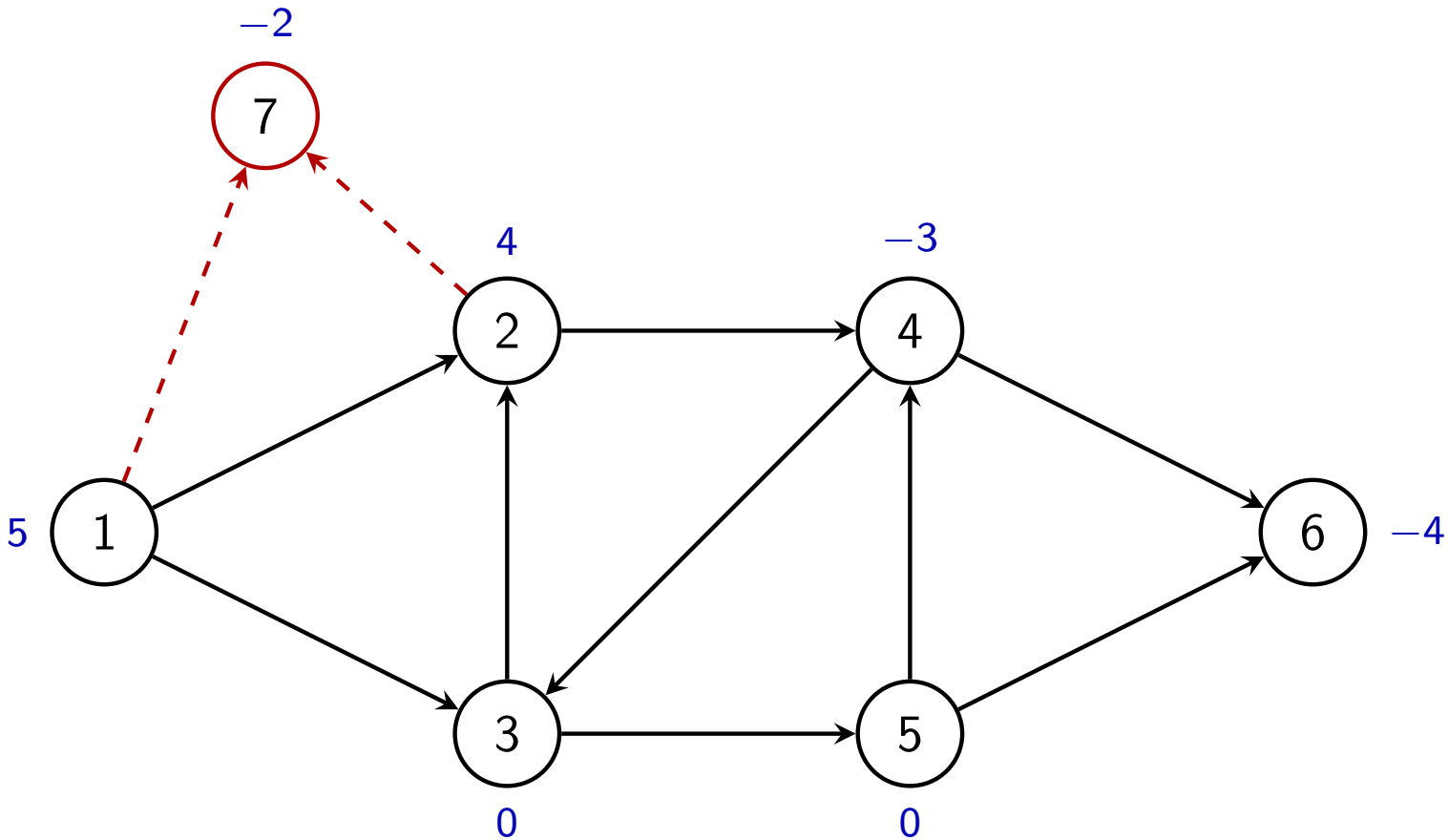
Example: a dummy node on the reference network (1/2)

- In the reference network we raise the supply of node 2 to $b_2 = 4$
- The total supply is 9, the total demand is 7, and the excess is $e = 2$
- The balanced network adds node 7 with $b_7 = -2$, and the arcs $(1, 7)$ and $(2, 7)$ with cost 0 and capacity $+\infty$
- Its optimal flow, computed with Gurobi, ships all the 4 units of node 2 and only 3 units of node 1:

$$x_{17} = 2, \qquad x_{27} = 0, \qquad \text{with optimal value } 56$$

Example: a dummy node on the reference network (2/2)

The balanced network: the dummy node 7 and the dummy arcs (dashed), with the balances in blue.



Summary

- An unbalanced instance becomes a balanced network with a **dummy node** that absorbs the excess
- The dummy arcs cost zero, so the cost of the flow does not change
- The flow on a dummy arc is the unused supply of its tail

Total unimodularity and integrality

Introduction: Total unimodularity and integrality

- In many applications the flow must be **integer**: vehicles, people, containers
- We prove that the incidence matrix is totally unimodular, and that therefore every vertex of the feasible region is integer
- Consequence: the MCF model solved as an LP already gives an integer optimal flow, and the integrality constraints come for free

Definition 8: totally unimodular matrix

An integer matrix $\boldsymbol{M}$ *is* ***totally unimodular*** *(TU) if the determinant of every square submatrix of* $\boldsymbol{M}$ *is* 0, $+1$ *or* -1. *A square submatrix is obtained by choosing* k *rows and* k *columns, for any* $k \geq 1$.

- With $k = 1$ the definition requires every entry of a TU matrix to be 0, $+1$ or -1

Theorem 1: the incidence matrix is totally unimodular

The node–arc incidence matrix $\boldsymbol{B}$ *of every directed graph* $G = (N, A)$ *is totally unimodular.*

- Idea of the proof: induction on the size k of the submatrix, using that every column of $\boldsymbol{B}$ has exactly one $+1$ and one -1

Proof (1/2)

- Let $\boldsymbol{M}$ be a $k \times k$ submatrix of $\boldsymbol{B}$; we prove $\det(\boldsymbol{M}) \in \{0, +1, -1\}$ by induction on k
- For $k = 1$ the entry is 0, $+1$ or -1
- Let $k \geq 2$. Every column of $\boldsymbol{B}$ has exactly one $+1$ and one -1, so every column of $\boldsymbol{M}$ has **at most two** nonzero entries
- There are three cases

Proof (2/2)

1. *A column of* $\boldsymbol{M}$ *is zero.* Then $\det(\boldsymbol{M}) = 0$.

2. *A column of* $\boldsymbol{M}$ *has exactly one nonzero entry,* ± 1*.* The Laplace expansion along this column gives $\det(\boldsymbol{M}) = \pm \det(\boldsymbol{M}')$, where $\boldsymbol{M}'$ is a $(k-1) \times (k-1)$ submatrix of $\boldsymbol{B}$: by induction $\det(\boldsymbol{M}') \in \{0, +1, -1\}$.

3. *Every column of* $\boldsymbol{M}$ *has two nonzero entries,* $+1$ *and* -1*.* Then the rows of $\boldsymbol{M}$ add up to the zero vector, they are linearly dependent, and $\det(\boldsymbol{M}) = 0$.

□

Example: submatrices of the reference incidence matrix (1/2)

We take rows $2, 3, 4$ of the incidence matrix $\boldsymbol{B}$ of the reference graph and two choices of three columns.

	a_1	a_3	a_5
2	-1	1	0
3	0	0	1
4	0	-1	0

	a_3	a_4	a_6
2	1	-1	0
3	0	1	-1
4	-1	0	1

- The column of $a_1 = (1, 2)$ in the first matrix has one nonzero entry (case 2): expanding along it, and then along the column of a_5, the determinant is -1

Example: submatrices of the reference incidence matrix (2/2)

- The arcs $a_3 = (2, 4)$, $a_4 = (3, 2)$, $a_6 = (4, 3)$ of the second matrix form the cycle $2, 4, 3, 2$
- All their endpoints are among the chosen rows: every column has $+1$ and -1 (case 3), and the determinant is 0
- The matrix $\boldsymbol{B}$ has 5004 square submatrices; computing all their determinants confirms that they take only the values -1, 0 and $+1$

Integer vertices

- A point $\bar{\boldsymbol{x}}$ of a polyhedron P is a **vertex** if it is not the midpoint of two distinct points of P, as in the chapter «The geometry of Linear Programming» of the module *Linear Programming*
- For a set of arcs F, we denote by $\boldsymbol{B}_F$ the submatrix of the columns of F, and by $\boldsymbol{x}_F$ the corresponding entries of $\boldsymbol{x}$
- We now show that total unimodularity makes every vertex of the MCF polyhedron integer

Theorem 2: integer vertices

Let $G = (N, A)$ be a network with integer balances $\boldsymbol{b}$, integer lower bounds $\boldsymbol{l}$ and capacities $\boldsymbol{u}$ that are integer or $+\infty$. Then every vertex of

$$P = \left\{ \boldsymbol{x} \in \mathbb{R}^m : \boldsymbol{B}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u} \right\}$$

is an integer vector.

- Idea of the proof: the entries strictly between their bounds solve a nonsingular system with integer data, and Cramer's rule divides by $\det = \pm 1$

Proof (1/3)

- Let $\bar{\boldsymbol{x}}$ be a vertex of P, and let

$$F = \{\, a \in A : l_a < \bar{x}_a < u_a \,\}$$

  be the arcs strictly between their bounds
- If $F = \emptyset$, every entry of $\bar{\boldsymbol{x}}$ is a bound, hence an integer

1. *The columns of $\boldsymbol{B}_F$ are linearly independent.* Otherwise $\boldsymbol{B}_F \boldsymbol{d}_F = 0$ for some $\boldsymbol{d} \neq 0$ with $d_a = 0$ for $a \notin F$. For a small $\varepsilon > 0$ both $\bar{\boldsymbol{x}} + \varepsilon \boldsymbol{d}$ and $\bar{\boldsymbol{x}} - \varepsilon \boldsymbol{d}$ satisfy the strict bounds on F and $\boldsymbol{B}\boldsymbol{x} = \boldsymbol{b}$, so they lie in P, and $\bar{\boldsymbol{x}}$ is their midpoint: a contradiction.

Proof (2/3)

2. *The system for $\bar{\boldsymbol{x}}_F$ has integer data.* Every $a \notin F$ has $\bar{x}_a \in \{l_a, u_a\}$, an integer, so

$$\boldsymbol{B}_F \bar{\boldsymbol{x}}_F = \boldsymbol{b} - \sum_{a \notin F} \boldsymbol{B}_a \bar{x}_a =: \hat{\boldsymbol{b}}$$

   is integer.

3. *Cramer's rule.* $\boldsymbol{B}_F$ has rank $|F|$, so some $|F|$ of its rows form a nonsingular square matrix $\boldsymbol{M}$, and $\boldsymbol{M}\bar{\boldsymbol{x}}_F = \hat{\boldsymbol{b}}_R$ on these rows R. By Cramer's rule each entry of $\bar{\boldsymbol{x}}_F$ is

$$\frac{\det(\boldsymbol{M}_a)}{\det(\boldsymbol{M})},$$

   with $\boldsymbol{M}_a$ an integer matrix.

Proof (3/3)

4. By the theorem on the incidence matrix, $\boldsymbol{B}$ is totally unimodular, so

$$\det(\boldsymbol{M}) = \pm 1,$$

and every entry of $\bar{\boldsymbol{x}}_F$ is an integer.

□

Corollary 1: integrality of the minimum cost flow model

Let the balances, the lower bounds and the capacities be integer, with every capacity finite. If the MCF *model has a feasible solution, then it has an integer optimal solution, for any costs* $\boldsymbol{c}$.

Proof

- The polyhedron P is nonempty and bounded, since $\boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u}$ with finite bounds
- A linear function attains its minimum over a nonempty bounded polyhedron at a vertex (chapter «The geometry of Linear Programming»)
- Every vertex of P is integer by the theorem on integer vertices

□

Integrality for free

- The simplex algorithm ends at a vertex, so with integer data it returns an integer optimal flow: the integer requirement $x_{ij} \in \mathbb{Z}_{\geq 0}$ **comes for free**
- Total unimodularity is a property of the **matrix**: for a matrix that is not TU, an integer right-hand side can give fractional vertices

Example: a matrix that is not totally unimodular

- The undirected triangle with vertices $1, 2, 3$ and edges $e_1 = \{1, 2\}$, $e_2 = \{1, 3\}$, $e_3 = \{2, 3\}$ has the vertex–edge incidence matrix

$$\boldsymbol{M} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad \det(\boldsymbol{M}) = -2,$$

  so $\boldsymbol{M}$ is not TU

- The polyhedron $\{\boldsymbol{y} \in \mathbb{R}^3 : \boldsymbol{M}\boldsymbol{y} = 1,\ \boldsymbol{y} \geq 0\}$ contains the single point

$$\boldsymbol{y} = \left(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}\right),$$

  since $\boldsymbol{M}$ is nonsingular

- This vertex is fractional, although the data are integer

Summary

- The incidence matrix of a directed graph is **totally unimodular**
- With integer balances and bounds, every vertex of the feasible region is integer
- The MCF model has an integer optimal solution, for any costs
- Without total unimodularity, integer data can give fractional vertices

The dual model and the reduced costs

Introduction: The dual model and the reduced costs

- The dual of the MCF model has one variable per node, the **potential** π_i
- The potentials define the **reduced costs** of the arcs
- With them we can **certify** that a flow is optimal, without a solver: this is the basis of the optimality tests of the algorithms for flows

Setting up the dual

- By the proposition on removing the lower bounds, the lower bounds can be removed: in this section $\boldsymbol{l} = 0$ and every capacity is finite
- The constraint $x_{ij} \leq u_{ij}$ is written as $-x_{ij} \geq -u_{ij}$
- Following the rules of the chapter «Duality theory» of the module *Linear Programming*, we use
  - a free dual variable π_i, the **potential** of node i, for its conservation equation,
  - a non-negative dual variable α_{ij} for the capacity of (i, j)

The $D(\mathrm{MCF})$ model

Using free dual variables π_i for the flow conservation constraints and non-negative dual variables α_{ij} for the capacities, the dual model of the MCF model with $l = 0$ is:

$$\max \sum_{i \in N} b_i \pi_i - \sum_{(i,j) \in A} u_{ij} \alpha_{ij}$$

$$\text{subject to} \quad \pi_i - \pi_j - \alpha_{ij} \leq c_{ij}, \quad \forall (i,j) \in A,$$

$$\pi_i \in \mathbb{R}, \qquad \forall i \in N,$$

$$\alpha_{ij} \geq 0, \qquad \forall (i,j) \in A.$$

This LP model is referred to as the $D(\mathrm{MCF})$ model.

Description of the dual model

- The column of x_{ij} has $+1$ in the equation of i, -1 in the equation of j and -1 in its capacity constraint: this gives the dual constraint $\pi_i - \pi_j - \alpha_{ij} \leq c_{ij}$, one for every arc
- The primal variables are non-negative, so the dual constraints are of type $\leq$
- Adding the same constant to all the potentials changes neither the dual constraints nor the objective, since $\sum_{i \in N} b_i = 0$: the potentials can be **normalized**, for example with $\pi_n = 0$

Definition 9: reduced cost

Given potentials $\boldsymbol{\pi} \in \mathbb{R}^n$, *the* ***reduced cost*** *of the arc* $(i, j) \in A$ *is*

$$\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j.$$

- The dual constraint $\pi_i - \pi_j - \alpha_{ij} \leq c_{ij}$ reads

$$\bar{c}_{ij} + \alpha_{ij} \geq 0$$

- For fixed potentials, the best choice of $\alpha_{ij} \geq 0$ is the smallest one,

$$\alpha_{ij} = \max\{0, -\bar{c}_{ij}\}$$

Lemma 2: cost and reduced cost

For every $\boldsymbol{x}$ *with* $\boldsymbol{Bx} = \boldsymbol{b}$ *and every* $\boldsymbol{\pi} \in \mathbb{R}^n$,

$$\sum_{(i,j)\in A} c_{ij}\, x_{ij} = \sum_{(i,j)\in A} \bar{c}_{ij}\, x_{ij} + \sum_{i\in N} b_i\, \pi_i .$$

Proof

- Since $c_{ij} = \bar{c}_{ij} + \pi_i - \pi_j$, it suffices to show

$$\sum_{(i,j)\in A} (\pi_i - \pi_j)\, x_{ij} = \sum_{i\in N} b_i \pi_i$$

- The term $\pi_i x_{ij}$ goes to the tail and $-\pi_j x_{ij}$ to the head, so

$$\sum_{(i,j)\in A} (\pi_i - \pi_j)\, x_{ij} = \sum_{i\in N} \pi_i \Big(\sum_{(i,j)\in \delta^+(i)} x_{ij} - \sum_{(j,i)\in \delta^-(i)} x_{ji} \Big)$$

$$= \sum_{i\in N} \pi_i\, b_i.$$

□

Theorem 3: duality gap

Let $\boldsymbol{x}$ *be a feasible flow* ($\boldsymbol{l} = 0$) *and* $(\boldsymbol{\pi}, \boldsymbol{\alpha})$ *a feasible solution of the* $D(\mathrm{MCF})$ *model. Then*

$$\sum_{(i,j)\in A} c_{ij} x_{ij} - \left(\sum_{i \in N} b_i \pi_i - \sum_{(i,j)\in A} u_{ij} \alpha_{ij} \right)$$
$$= \sum_{(i,j)\in A} (\bar{c}_{ij} + \alpha_{ij})\, x_{ij} + \sum_{(i,j)\in A} \alpha_{ij}\, (u_{ij} - x_{ij}) \;\geq\; 0,$$

and both solutions are optimal if and only if, for every arc, $(\bar{c}_{ij} + \alpha_{ij})\, x_{ij} = 0$ *and* $\alpha_{ij}(u_{ij} - x_{ij}) = 0$.

Proof (1/2)

- By the lemma on cost and reduced cost, the left-hand side is

$$\sum_{(i,j)\in A} \bar{c}_{ij} x_{ij} + \sum_{(i,j)\in A} u_{ij}\alpha_{ij}$$

- The identity follows from

$$u_{ij}\alpha_{ij} = \alpha_{ij}x_{ij} + \alpha_{ij}(u_{ij} - x_{ij})$$

- Every term is non-negative:
  - $\bar{c}_{ij} + \alpha_{ij} \geq 0$ by the dual constraint,
  - $\alpha_{ij} \geq 0$,
  - $0 \leq x_{ij} \leq u_{ij}$

Proof (2/2)

- If the gap is zero, the primal cost equals a dual value, which is a lower bound on every primal cost by the inequality just proved: $\boldsymbol{x}$ is optimal, and in the same way $(\boldsymbol{\pi}, \boldsymbol{\alpha})$ is optimal
- If both are optimal, the gap is zero by the strong duality theorem of the chapter «Duality theory»
- A sum of non-negative terms is zero if and only if every term is zero

□

Corollary 2: reduced cost certificate

Let $\boldsymbol{x}$ *be a feasible flow (*$\boldsymbol{l} = 0$*) and* $\boldsymbol{\pi} \in \mathbb{R}^n$ *potentials such that, for every arc,*

$$\bar{c}_{ij} > 0 \;\Rightarrow\; x_{ij} = 0, \qquad\qquad \bar{c}_{ij} < 0 \;\Rightarrow\; x_{ij} = u_{ij}.$$

Then $\boldsymbol{x}$ *is an optimal flow, and* $\boldsymbol{\pi}$ *with* $\alpha_{ij} = \max\{0, -\bar{c}_{ij}\}$ *is an optimal solution of the D*(MCF) *model.*

Proof

- The pair $(\boldsymbol{\pi}, \boldsymbol{\alpha})$ is dual feasible, since $\alpha_{ij} \geq 0$ and $\bar{c}_{ij} + \alpha_{ij} \geq 0$
- We check the two conditions of the duality gap theorem on every arc:
  - if $\bar{c}_{ij} \geq 0$, then $\alpha_{ij} = 0$, and $\bar{c}_{ij} x_{ij} = 0$ because $\bar{c}_{ij} > 0$ forces $x_{ij} = 0$;
  - if $\bar{c}_{ij} < 0$, then $\bar{c}_{ij} + \alpha_{ij} = 0$, and $\alpha_{ij}(u_{ij} - x_{ij}) = 0$ because $x_{ij} = u_{ij}$
- The gap is zero, so both solutions are optimal

□

Reading the certificate

An arc with **positive** reduced cost is **empty**,

an arc with **negative** reduced cost is **saturated**,

and an arc strictly between its bounds
has reduced cost **zero**.

Example: the dual of the reference network (1/2)

For the reference network, the objective of the $D(\mathrm{MCF})$ model is

$$\begin{aligned}\max\ & 5\pi_1 + 2\pi_2 - 3\pi_4 - 4\pi_6 \\ & - \big(4\alpha_{12} + 4\alpha_{13} + 4\alpha_{24} + 2\alpha_{32} + 4\alpha_{35} \\ & \quad + 3\alpha_{43} + 5\alpha_{46} + 3\alpha_{54} + 2\alpha_{56}\big)\end{aligned}$$

- The potentials are free and $\boldsymbol{\alpha} \geq 0$
- The balances $b_3 = b_5 = 0$ make π_3 and π_5 disappear from the objective

Example: the dual of the reference network (2/2)

One constraint per arc:

$$\begin{aligned}
\pi_1 - \pi_2 - \alpha_{12} &\leq 2, \\
\pi_1 - \pi_3 - \alpha_{13} &\leq 2, \\
\pi_2 - \pi_4 - \alpha_{24} &\leq 5, \\
\pi_3 - \pi_2 - \alpha_{32} &\leq 2, \\
\pi_3 - \pi_5 - \alpha_{35} &\leq 3, \\
\pi_4 - \pi_3 - \alpha_{43} &\leq 3, \\
\pi_4 - \pi_6 - \alpha_{46} &\leq 5, \\
\pi_5 - \pi_4 - \alpha_{54} &\leq 5, \\
\pi_5 - \pi_6 - \alpha_{56} &\leq 3.
\end{aligned}$$

Summary

The dual model:

- one potential π_i per node, one $\alpha_{ij} \geq 0$ per capacity
- the potentials are defined up to a constant, fixed by a normalization

Reduced costs and optimality:

- the reduced cost is $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$
- a feasible flow whose positive-reduced-cost arcs are empty and negative-reduced-cost arcs are saturated is optimal
- potentials and reduced costs are the basis of the optimality tests of the chapter «Minimum cost flow»

Special cases

Introduction: Special cases

- Several classical problems are MCF models with **special data**
- All the lower bounds are zero; only the balances, the capacities and the costs change
- Everything proved in this chapter, integrality and certificates, holds for all of them

Special cases (1/2)

problem	the MCF model
Shortest path	$b_s = 1$, $b_t = -1$, $b_i = 0$ otherwise; $u_{ij} = +\infty$, c_{ij} the length of the arc: the unit of flow travels from s to t.
Maximum flow	$b_i = 0$ and $c_{ij} = 0$ for every arc, plus the arc (t, s) with $c_{ts} = -1$ and $u_{ts} = +\infty$: the flow on (t, s) is the value sent from s to t.

Special cases (2/2)

problem	the MCF model
Transportation	$N = P \cup D$ with arcs from the origins P to the destinations D; b_i the supply of $i \in P$, $-b_j$ the demand of $j \in D$, $u_{ij} = +\infty$.
Assignment	a transportation model with $\|P\| = \|D\|$, $b_i = 1$ for $i \in P$ and $b_j = -1$ for $j \in D$.

- In the four models the balances and the bounds are integer, so the integer vertices of the theorem on integer vertices are paths, integer flows and assignments

Summary

- Shortest path, maximum flow, transportation and assignment are **minimum cost flow models** with special balances, capacities and costs
- Their data are integer, so their LP models have integer optimal vertices

A transshipment instance

Introduction: A transshipment instance

- We solve the MCF model of the reference network: two supply nodes, two demand nodes and two transshipment nodes
- The optimal solution is computed with Gurobi (gurobipy)
- Then we **certify it by hand** with the reduced cost certificate

Example: the optimal flow (1/2)

The optimal flow $\tilde{\boldsymbol{x}}$ and its optimal value are

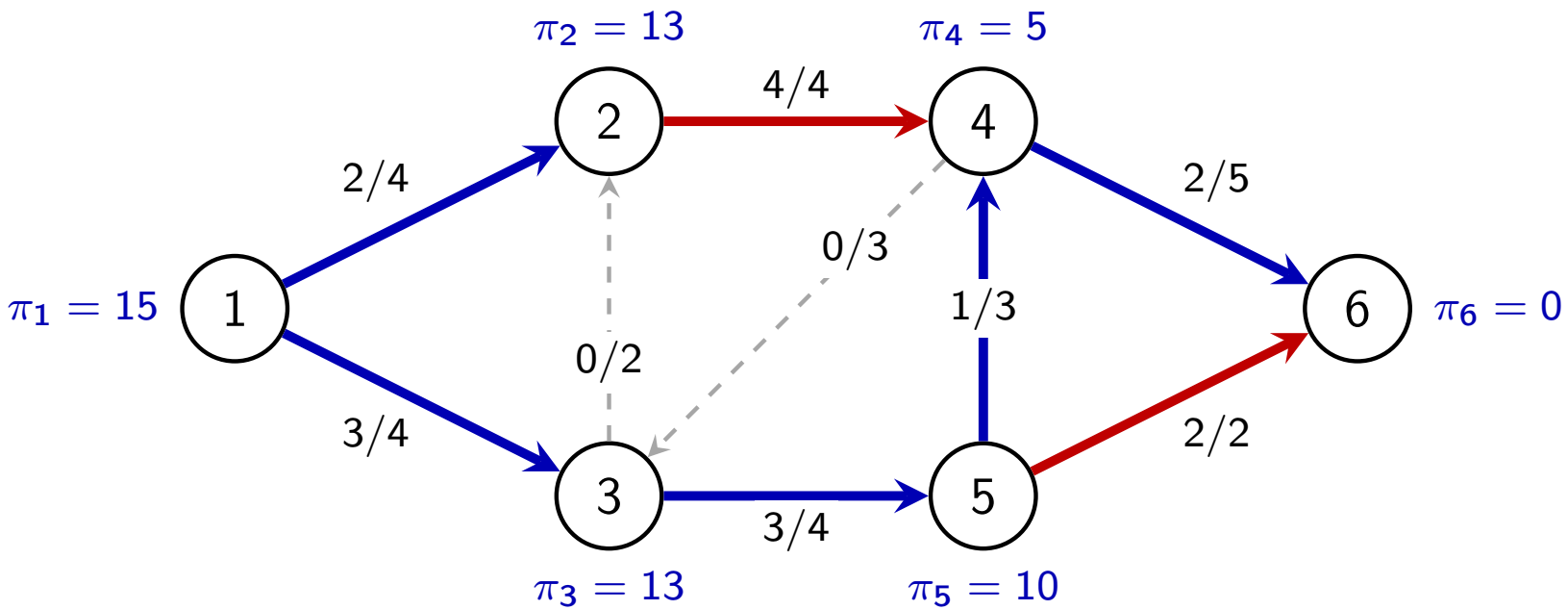
arc	$(1,2)$	$(1,3)$	$(2,4)$	$(3,2)$	$(3,5)$	$(4,3)$	$(4,6)$	$(5,4)$	$(5,6)$
$\tilde{x}_{ij}$	2	3	4	0	3	0	2	1	2
u_{ij}	4	4	4	2	4	3	5	3	2

$$z(\mathrm{MCF}) = 60$$

- The flow is integer, as the corollary on the integrality of the minimum cost flow model guarantees

Example: the optimal flow (2/2)

Each arc carries $\tilde{x}_{ij}/u_{ij}$: the arcs strictly between their bounds are blue, the arcs at capacity red, the empty arcs grey and dashed. Next to each node, its potential.



The certificate of optimality

1. The five blue arcs $(1,2)$, $(1,3)$, $(3,5)$, $(4,6)$, $(5,4)$ form a spanning tree. We ask for reduced cost zero on them, $\pi_i - \pi_j = c_{ij}$, and normalize with $\pi_6 = 0$.

2. Going backwards from node 6:

$$\pi_4 = \pi_6 + c_{46} = 5, \quad \pi_5 = \pi_4 + c_{54} = 10,$$
$$\pi_3 = \pi_5 + c_{35} = 13, \quad \pi_1 = \pi_3 + c_{13} = 15,$$
$$\pi_2 = \pi_1 - c_{12} = 13.$$

3. The potential π_i is the cost of the tree path from i to 6: for example,

$$\pi_1 = c_{13} + c_{35} + c_{54} + c_{46} = 2 + 3 + 5 + 5 = 15.$$

Example: reduced costs and optimality (1/2)

The reduced costs $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$ are:

arc	$(1,2)$	$(1,3)$	$(2,4)$	$(3,2)$	$(3,5)$	$(4,3)$	$(4,6)$	$(5,4)$	$(5,6)$
$\bar{c}_{ij}$	0	0	-3	2	0	11	0	0	-7
α_{ij}	0	0	3	0	0	0	0	0	7

- The arcs with $\bar{c}_{ij} > 0$, $(3,2)$ and $(4,3)$, are empty
- The arcs with $\bar{c}_{ij} < 0$, $(2,4)$ and $(5,6)$, are at capacity
- By the reduced cost certificate, $\tilde{\boldsymbol{x}}$ is optimal

Example: reduced costs and optimality (2/2)

- The dual value is

$$\sum_{i \in N} b_i \pi_i - \sum_{(i,j) \in A} u_{ij} \alpha_{ij} = 86 - 26 = 60,$$

  equal to the cost of $\tilde{\boldsymbol{x}}$

- The potentials coincide with the dual values returned by Gurobi, shifted so that $\pi_6 = 0$

Uniqueness of the optimal flow

- Every optimal flow has gap zero with $(\boldsymbol{\pi}, \boldsymbol{\alpha})$, so by the duality gap theorem it has

$$x_{32} = x_{43} = 0, \qquad x_{24} = 4, \qquad x_{56} = 2$$

- The flow on the five tree arcs is then fixed by conservation, since their columns are linearly independent

- Hence $\tilde{\boldsymbol{x}}$ is the **unique** optimal flow

The gap of the flow $\bar{x}$

- The feasible flow $\bar{x}$ of the reference network costs 63, and its gap with the same dual solution is

$$63 - 60 = 3 = \alpha_{24}(u_{24} - \bar{x}_{24}) = 3 \cdot (4 - 3) :$$

the arc $(2, 4)$ is not used enough

Summary

- On the reference network the optimal flow costs 60
- The potentials of its spanning tree give reduced costs that are positive only on empty arcs and negative only on saturated arcs
- They **certify the optimality** without a solver, and show that the optimal flow is unique

Typical mistakes

Introduction: Typical mistakes

- We collect the mistakes that are made most often when writing and using the network flow models
- Each of them comes from a definition or a result of this chapter
- Check them before solving the exercises

Typical mistakes (1/4)

Observation 1: Swapping outflow and inflow

The conservation constraint is outflow minus inflow equal to b_i, with $b_i > 0$ at a supply node: the signs follow the incidence matrix, $+1$ in the tail.

Observation 2: Taking $\sum_i b_i = 0$ as sufficient

A balanced network can have no feasible flow: a cut may be too small to carry the supply inside it.

Typical mistakes (2/4)

Observation 3: Getting the substitution signs wrong

With $\boldsymbol{x} = \boldsymbol{l} + \boldsymbol{x}'$ the tail of (i, j) loses l_{ij} units of balance and the head gains them; the constant $\boldsymbol{c}'\boldsymbol{l}$ must be added back to the value.

Observation 4: Forgetting the dummy node

An instance with more supply than demand has no feasible flow with equalities; the dummy arcs must cost zero, or the unused supply is charged.

Typical mistakes (3/4)

Observation 5: Claiming integrality for any matrix

Integer vertices need a TU matrix and integer $\boldsymbol{b}$, $\boldsymbol{l}$, $\boldsymbol{u}$; *the costs may be fractional.*

Observation 6: Writing the reduced cost with the wrong signs

In this module $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$, *with* $-\pi$ *of the tail and* $+\pi$ *of the head.*

Typical mistakes (4/4)

Observation 7: Expecting unique potentials

The potentials are defined up to an additive constant; they are fixed by a normalization such as $\pi_n = 0$.

Summary

Writing the models:

- outflow minus inflow equals b_i, with $+1$ in the tail
- the substitution moves l_{ij} from the tail to the head, and adds back $\boldsymbol{c}'\boldsymbol{l}$
- an excess of supply needs a dummy node with zero-cost arcs

Using the results:

- the balance is necessary, not sufficient
- integrality needs a TU matrix and integer $\boldsymbol{b}$, $\boldsymbol{l}$, $\boldsymbol{u}$
- $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$, potentials up to a constant

In summary

Introduction: In summary

- We collect the key ideas of the chapter on network flow models
- The last table gathers the numbers of the reference network

Summary (1/3)

- A network has balances b_i, lower bounds l_{ij}, capacities u_{ij} and costs c_{ij}; the minimum cost flow problem is the LP

$$\min\{\boldsymbol{c}'\boldsymbol{x} : \boldsymbol{B}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{l} \leq \boldsymbol{x} \leq \boldsymbol{u}\}$$

- Across every cut the net outflow equals the balance inside; a feasible network is balanced and satisfies the cut condition
- Lower bounds are removed with $\boldsymbol{x} = \boldsymbol{l} + \boldsymbol{x}'$; an excess of supply goes to a dummy node through zero-cost arcs

Summary (2/3)

- The incidence matrix is totally unimodular, so with integer $\boldsymbol{b}$, $\boldsymbol{l}$, $\boldsymbol{u}$ every vertex is integer and the MCF model has an integer optimal solution
- The dual has potentials π_i; with $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$, a flow with empty positive-cost arcs and saturated negative-cost arcs is optimal
- Shortest path, maximum flow, transportation and assignment are MCF models

Summary (3/3)

The reference network at a glance.

balances	$\boldsymbol{b} = (5, 2, 0, -3, 0, -4)$
optimal flow	$\tilde{\boldsymbol{x}} = (2, 3, 4, 0, 3, 0, 2, 1, 2)$, $z(\mathrm{MCF}) = 60$
potentials ($\pi_6 = 0$)	$\boldsymbol{\pi} = (15, 13, 13, 5, 10, 0)$
empty / saturated arcs	$(3, 2)$, $(4, 3)$ / $(2, 4)$, $(5, 6)$
with $l_{32} = 1$	$\boldsymbol{b}' = (5, 3, -1, -3, 0, -4)$, optimal value 62
with $b_2 = 4$ (dummy node 7)	$x_{17} = 2$, optimal value 56